

Chapter 2

Dual space: bras, kets, adjoint operator

2.1 The idea of duality

Duality is the guiding thread of this entire lesson. Let us try to give it an intuitive interpretation. In mathematics, the notion of duality covers many situations, but in the setting of Hilbert spaces it expresses a simple and deep idea: *a vector can be viewed not only as an object in its own right, but also as an operator acting on other vectors to produce numbers* (that is, as a linear form). In quantum mechanics, the number thus produced is a probability amplitude.

The quantum formalism thus fully exploits these two facets, and the *Dirac notation* provides an ingenious system of writing that allows them to be manipulated with ease. It features *kets*, written $|u\rangle$, which represent vectors of the space, and *bras*, written $\langle u|$, which represent linear forms acting on these vectors. Their combination $\langle u|v\rangle$ produces a scalar, called a *bracket*, while expressions of the form $\sum a_{ij} |u_i\rangle \langle v_j|$ describe operators.

This notation has become the standard language of quantum physics, and full mastery of it is indispensable. A summary formula sheet of the calculation rules will be provided at the end of this lesson. In practice, it is perfectly possible to learn to manipulate these symbols without really understanding the mathematics hidden behind them. But to grasp their deep meaning, one must tackle the program we outline here: understanding how the complete identification between vectors and linear forms is constructed, and why this approach naturally leads to the definition of another essential object: the adjoint of an operator.

In the next section, we will try to establish such an isomorphism between a vector space and its algebraic dual, defined as the set of *all* its linear forms. We will see that this is impossible. Indeed, although a finite-dimensional vector space is in fact isomorphic to its dual, this isomorphism is not canonical (it depends on an arbitrary choice of basis). Moreover, the situation worsens in infinite dimension: such an isomorphism simply does not exist. These obstacles invalidate the automatic identification between vectors and linear forms, and justify appealing to the richer structure of Hilbert spaces.

Section 3 shows how to exploit this additional structure to overcome these difficulties. Thanks to the inner product, and by restricting attention to the continuous linear forms

that make up what is called the topological dual, one reaches a remarkable result. The Riesz theorem asserts that for any Hilbert space of arbitrary dimension, there exists a canonical antilinear isometric isomorphism between the space and its topological dual.

This result rigorously grounds Dirac notation, which we will then detail in Sections 4 and 5 (kets and bras, and linear operators), before showing how the Riesz isomorphism naturally gives rise to the notion of the adjoint of an operator (Section 6). Section 7 summarizes this construction in a complete diagram. Throughout this lesson, we will use the finite-dimensional case as a concrete example to illustrate the theory, systematically working over the field $\mathbb{C}$.

2.2 Linear forms and the algebraic dual

Let E be a vector space over $\mathbb{C}$. We define its *algebraic dual* E^* as the set of **linear forms** on E , i.e. linear maps $\varphi : E \rightarrow \mathbb{C}$.

In finite dimension n , fixing a basis $(e_1, \dots, e_n)$ of E , one can construct a corresponding family of linear forms $(\theta_1, \dots, \theta_n)$ in E^* , defined by the relation:

$$\theta_j(e_i) = \delta_{ij}$$

One then shows that this family is linearly independent and generating, and therefore a basis of E^* (called the dual basis). One deduces that $\dim(E^*) = \dim(E)$, which implies that E and E^* are isomorphic. A possible isomorphism is the map T that associates with each vector $x = \sum x_i e_i$ the linear form $w_x = \sum x_i \theta_i$.

However, this isomorphism T is, so to speak, artificial, in the sense that it depends on the initial choice of a basis of E . If one changes basis, the map T also changes. Hence there is no canonical (natural) identification between a vector space and its algebraic dual.

In infinite dimension, a classical result shows that E is never isomorphic to its algebraic dual, since the latter has a strictly larger dimension (cf. the Erdős-Kaplansky theorem). For example, if E has countable algebraic dimension, its algebraic dual has uncountable algebraic dimension.

In both cases, then, there is no canonical identification between vectors and linear forms.

2.3 Topological dual and the Riesz theorem

In a Hilbert space $\mathcal{H}$, the norm associated with the inner product allows one to single out a particular class of linear forms: those that are *continuous*. One says that $\varphi : \mathcal{H} \rightarrow \mathbb{C}$ is continuous at x_0 if

$$\forall \varepsilon > 0, \exists \delta > 0 : \|x - x_0\|_{\mathcal{H}} < \delta \implies |\varphi(x) - \varphi(x_0)| < \varepsilon,$$

and if it is continuous at a point, then by linearity it is continuous everywhere¹. The set of continuous linear forms is a vector subspace of the algebraic dual. It is called the topological dual and is still denoted (by abuse of notation) $\mathcal{H}^*$.

¹We will revisit these points in the following lessons (topology, theory of linear operators).

Definition 2.1 : Topological dual

Let $\mathcal{H}$ be a Hilbert space over $\mathbb{C}$. The set of continuous linear forms on $\mathcal{H}$ is a vector space, called the *topological dual* of $\mathcal{H}$ and denoted $\mathcal{H}^*$. This space carries a natural norm, called the dual norm and given by

$$\|\varphi\|_{\mathcal{H}^*} \stackrel{\text{def}}{=} \sup_{\|x\|_{\mathcal{H}}=1} |\varphi(x)|,$$

for which $\mathcal{H}^*$ is a complete normed vector space (i.e. a Banach space).

Remark: this norm is a special case of the operator norm that will appear later (cf. the lesson on linear operators). In itself, the above construction is valid for any normed space. What makes the Hilbert case special is the following theorem: one uses the inner product to construct a bijective identification of every continuous linear form with a vector:

Theorem 2.1 : Riesz representation theorem

Let $\mathcal{H}$ be a Hilbert space, separable or not. Every continuous linear form $\varphi \in \mathcal{H}^*$ can be written in a *unique* way as

$$\varphi(x) = \langle u, x \rangle$$

for some vector $u \in \mathcal{H}$.

We note that the map $\Phi : u \mapsto \varphi_u = \langle u, \cdot \rangle$ from $\mathcal{H}$ to $\mathcal{H}^*$ is injective even in a space that is merely pre-Hilbert. Indeed, if $\varphi_u = \varphi_v$, then for every x in $\mathcal{H}$ we have $0 = \varphi_u(x) - \varphi_v(x) = \langle u - v, x \rangle$, and hence $u = v$ by choosing $x = u - v$ and using the positive definiteness of the inner product.

The Riesz theorem then asserts that this map is also surjective. This is not trivial, and in fact only holds for Hilbert spaces. For instance, in an incomplete pre-Hilbert space the map is not surjective, since there exist continuous linear forms that cannot be written as an inner product with a vector of the incomplete space.

The Riesz theorem thus provides the bijection Φ between $\mathcal{H}$ and $\mathcal{H}^*$ that we were seeking. We deduce the following corollary:

Corollary 2.1

The map $\Phi : u \mapsto \varphi_u$ is an antilinear isometric isomorphism, called the *canonical Riesz isomorphism* between $\mathcal{H}$ and its topological dual endowed with its dual norm: $\mathcal{H}^* \simeq \mathcal{H}$.

Proof. 1. Antilinearity follows from that of the inner product on $\mathcal{H}$: $\Phi(\lambda u) = \varphi_{\lambda u} = \langle \lambda u, \cdot \rangle = \lambda^* \langle u, \cdot \rangle = \lambda^* \varphi_u = \lambda^* \Phi(u)$.

2. Let us show the isometry, i.e. that $\|\Phi(u)\|_{\mathcal{H}^*} = \|u\|_{\mathcal{H}}$. We first have:

$$\|\varphi_u\|_{\mathcal{H}^*} = \sup_{\|x\|=1} |\langle u, x \rangle| \leq \|u\|_{\mathcal{H}}$$

by the Cauchy-Schwarz inequality. One then notes that equality is attained for $x = u/\|u\|$ if $u \neq 0$. Equality is trivial if $u = 0$.

3. We use the map Φ , an isometric bijection, to transport the inner product of $\mathcal{H}$ onto $\mathcal{H}^*$ by defining:

$$\langle \varphi_u, \varphi_v \rangle_{\mathcal{H}^*} \stackrel{\text{def}}{=} \langle \Phi^{-1}(\varphi_v), \Phi^{-1}(\varphi_u) \rangle_{\mathcal{H}} = \langle v, u \rangle_{\mathcal{H}}.$$

Note the reversed order (v, u instead of u, v) to compensate for the antilinearity of Φ . Let us indeed verify the sesquilinearity of this inner product on $\mathcal{H}^*$: $\langle \lambda \varphi_u, \varphi_v \rangle_{\mathcal{H}^*} = \langle \varphi_{\lambda^* u}, \varphi_v \rangle_{\mathcal{H}^*} = \langle v, \lambda^* u \rangle_{\mathcal{H}} = \lambda^* \langle v, u \rangle_{\mathcal{H}} = \lambda^* \langle \varphi_u, \varphi_v \rangle_{\mathcal{H}^*}$. The other inner product axioms are easily verified.

4. The norm induced by this inner product satisfies:

$$\sqrt{\langle \varphi_u, \varphi_u \rangle_{\mathcal{H}^*}} = \sqrt{\langle u, u \rangle_{\mathcal{H}}} = \|u\|_{\mathcal{H}} = \|\varphi_u\|_{\mathcal{H}^*},$$

and therefore coincides with the dual norm.

5. Finally, the completeness of $\mathcal{H}^*$, and hence its Hilbert space character, is guaranteed by the fact that a bijective isometry from a complete space preserves completeness, which we take for granted here.

Thus $\mathcal{H}^*$ inherits a Hilbert space structure for which Φ becomes an (antilinear) isomorphism of Hilbert spaces. $\square$

2.4 Dirac notation: bras, kets, brackets

Dirac notation is nothing other than a rewriting of the Riesz isomorphism. We first define kets, bras, and brackets. We then use them to rewrite the decomposition on a Hilbert basis seen in the previous lesson.

1. **Kets.** A vector $u \in \mathcal{H}$ is written in the form of a **ket**, denoted as follows:

$$\boxed{u \in \mathcal{H} \quad \longleftrightarrow \quad |u\rangle}$$

The linear structure gives us the following calculation rules:

$$|u + v\rangle = |u\rangle + |v\rangle \tag{2.1}$$

$$|\lambda u\rangle = \lambda |u\rangle \tag{2.2}$$

2. **Bras.** To every vector $u \in \mathcal{H}$ we associate, via Φ , a form φ_u . We write this linear form as a **bra**, placing the associated vector inside the bra:

$$\boxed{\varphi_u = \Phi(u) \in \mathcal{H}^* \quad \longleftrightarrow \quad \langle u|}$$

Thus $\langle u|$ represents the vector u seen as a linear form, i.e. the action "take the inner product with u and return a number". We have the following calculation rules:

$$\langle u + v| = \langle u| + \langle v| \tag{2.3}$$

$$\langle \lambda u| = \lambda^* \langle u|, \tag{2.4}$$

as a consequence of the antilinearity of Φ .

3. **The bracket.** The two notational devices above allow one to form an inner product, which is called a *bracket* in English, as a product of a bra with a ket:

$$\langle u, x \rangle \stackrel{\text{def}}{=} \langle u | x \rangle$$

This notation in fact reflects the action of the linear form $\langle u |$ on the vector $|x\rangle$:

$$\langle u | (|x\rangle) = \varphi_u(x) = \langle u, x \rangle = \langle u | x \rangle.$$

4. **Hilbert decomposition.** Regardless of the dimension of the Hilbert space, we have seen the decomposition formula $u = \sum_{i \in I} \langle e_i, u \rangle e_i$, where the sum is finite or converges in $\mathcal{H}$ if I is infinite. In Dirac notation it is written:

$$|u\rangle = \sum_{i \in I} \langle e_i | u \rangle |e_i\rangle = \sum_{i \in I} u_i |e_i\rangle \quad (2.5)$$

where the u_i are the components of the ket u in the basis. Note that they are obtained by orthogonal projection onto e_i in the form $u_i = \langle e_i | u \rangle$, and *not* in the form $\langle u | e_i \rangle$, which equals u_i^* . This is a common mistake, probably linked to the fact that in ordinary vector calculus on $\mathbb{R}^n$, one recovers the component v_i of a vector $\vec{v}$ via $v_i = \vec{v} \cdot \vec{e}_i$, which might suggest $u_i = \langle u | e_i \rangle$ in the quantum case. But this would ignore the sesquilinearity of the inner product on $\mathbb{C}$ and would immediately lead to a computational error.

For bras, we have:

$$\langle u | = \sum_{i \in I} \langle u | e_i \rangle \langle e_i | = \sum_{i \in I} u_i^* \langle e_i | \quad (2.6)$$

and the squared norm reads

$$\|u\|^2 = \langle u | u \rangle = \sum_{i \in I} |u_i|^2 = \sum_{i \in I} u_i^* u_i \quad (2.7)$$

where the second equality is Parseval's identity.

5. **The "dagger" operation.** Applying the isomorphism Φ (from kets to bras) is usually denoted by the superscript symbol $\dagger$. We choose to write the inverse isomorphism Φ^{-1} from bras to kets using the same symbol. This makes the dagger operation involutive. By notational convention, we therefore have:

$$\langle u | = |u\rangle^\dagger \quad (\text{map } \Phi) \quad (2.8)$$

$$|u\rangle = \langle u |^\dagger \quad (\text{map } \Phi^{-1}) \quad (2.9)$$

$$\left(|u\rangle^\dagger\right)^\dagger = |u\rangle \quad (\text{involution}) \quad (2.10)$$

Let us illustrate these various points in finite dimension. Take $\mathcal{H}$ of dimension n with a Hilbert basis $\mathcal{B} = (|e_i\rangle)_{i=1}^n$ equipped with its canonical inner product (cf. Lesson 1, the Hilbert space $\mathbb{C}^n$). Representing the basis kets canonically as column vectors, i.e.

as $(n, 1)$ matrices:

$$|e_i\rangle = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}_{\mathcal{B}}$$

where the 1 is in the i -th position, every ket $|u\rangle$ is written as a column vector

$$|u\rangle = \sum_{i=1}^n u_i |e_i\rangle = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix}_{\mathcal{B}} \in \mathbb{C}^n,$$

with $u_i = \langle e_i | u \rangle$. The linear form φ_u associated with the vector u must satisfy, for every vector v :

$$\varphi_u(v) = \langle u, v \rangle = \sum_{i=1}^n u_i^* v_i, \quad (\text{canonical inner product on } \mathbb{C}^n)$$

To obtain this sum, $\langle u |$ must therefore be represented by the $(1, n)$ row vector of the complex conjugate coordinates of u :

$$\langle u | = (u_1^*, u_2^*, \dots, u_n^*),$$

since then the inner product $\langle u | v \rangle$ is obtained as the usual matrix product:

$$\langle u | v \rangle = (u_1^*, u_2^*, \dots, u_n^*) \cdot \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} = \sum_{i=1}^n u_i^* v_i \in \mathbb{C}.$$

Thus, for example, in dimension 2:

$$|u\rangle = \begin{pmatrix} 1 \\ i \end{pmatrix} \Rightarrow \langle u | = (1, -i)$$

We have obtained the result:

Proposition 2.1 : Conjugate transpose

In finite dimension, the bra $\langle u |$ is the *conjugate transpose* of the ket $|u\rangle$. We have:

$$\langle u | = |u\rangle^\dagger = |u^*\rangle^\top \quad (2.11)$$

$$|u\rangle = \langle u |^\dagger = \langle u^* |^\top \quad (2.12)$$

Caution: this is meaningless in infinite dimension, because the transpose is not defined. That said, in the space $\ell^2(\mathbb{N})$ and in its canonical basis, everything proceeds in a relatively similar way, considering infinite column or row vectors and replacing finite sums with convergent series. This can be useful for building intuition, but strictly speaking these are neither matrices nor transpositions.

2.5 Operators, matrix elements, resolution of the identity

We continue our presentation of Dirac notation, this time also introducing linear maps (also called linear operators) $\hat{A}$ from $\mathcal{H}$ to $\mathcal{G}$, where $\mathcal{G}$ is another Hilbert space.

Writing operators. Note first that it is standard in physics to denote them with a hat. An operator $\hat{A}$ acting on a vector v yields the vector $\hat{A}v$. In Dirac notation, we therefore have two equivalent ways of writing this, the second being the more commonly used:

$$v = \hat{A}u \quad \longleftrightarrow \quad |v\rangle = |\hat{A}u\rangle \stackrel{\text{def}}{=} \hat{A}|u\rangle.$$

In the same way, an inner product involving an operator can be written as:

$$\langle w, \hat{A}u \rangle \quad \longleftrightarrow \quad \langle w|\hat{A}u\rangle \stackrel{\text{def}}{=} \langle w|\hat{A}|u\rangle$$

where the second notation is more frequent for aesthetic reasons.

Matrix elements. In the formula above, the case $\langle w| = \langle e_i|$ and $|u\rangle = |e_j\rangle$ is particularly important, as it defines what are called the matrix elements of the operator $\hat{A}$:

$$A_{ij} = \langle e_i|\hat{A}|e_j\rangle. \quad (2.13)$$

In infinite dimension, this expression does not always make sense, since $|e_i\rangle$ must still lie in the domain of definition of the operator $\hat{A}$. We will return to this. In finite dimension, on the other hand, this definition is straightforward: through the representation of basis vectors as column vectors, a linear operator is in one-to-one correspondence with its representing matrix; we therefore write $\hat{A} \longleftrightarrow A = (A_{ij})_{1 \leq i, j \leq n}$ with $A_{ij} = \langle e_i|\hat{A}|e_j\rangle$. The equation $|v\rangle = \hat{A}|u\rangle$ above then reads as the usual matrix product $\hat{A}|u\rangle = \sum_i \sum_j (A_{ij}u_j)|e_i\rangle$, while an inner product reads $\langle w|\hat{A}|u\rangle = (w^*)^\top Au = \sum_{ij} w_i^* A_{ij}u_j \in \mathbb{C}$.

Example 2.1 : Matrix computations

Explicitly, consider for example:

$$\hat{A} = \begin{pmatrix} i & 0 \\ 1 & -1 \end{pmatrix}, \quad |u\rangle = \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \quad |w\rangle = \begin{pmatrix} i \\ 1 \end{pmatrix},$$

then we have:

$$|v\rangle = \hat{A}|u\rangle = \begin{pmatrix} i & 0 \\ 1 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} i \\ -2 \end{pmatrix}, \quad \langle w|\hat{A}|u\rangle = (-i, 1) \cdot \begin{pmatrix} i & 0 \\ 1 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 3 \end{pmatrix} = -1.$$

Ket-bra operators. A bracket is a number, but a ket-bra is an operator, called an outer product (unrelated to the exterior product of differential forms). Given two vectors $|u\rangle$ and $|v\rangle$ of the *same* Hilbert space $\mathcal{H}$, one associates the operator $|u\rangle\langle v|$ from $\mathcal{H}$ to $\mathcal{H}$ defined by:

$$\begin{aligned} |u\rangle\langle v| &: \mathcal{H} \rightarrow \mathcal{H}, \\ |x\rangle &\mapsto |u\rangle \underbrace{\langle v|x\rangle}_{\in \mathbb{C}} = \langle v|x\rangle |u\rangle. \end{aligned}$$

Here too, *in finite dimension*, the case $\hat{E}_{ij} \stackrel{\text{def}}{=} |e_i\rangle \langle e_j|$ is particularly important. It is the operator whose matrix elements are all zero except for a "1" in row i and column j ; its associated matrix is therefore:

$$E_{ij} = \begin{pmatrix} 0 & \cdots & 0 & \cdots & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & \cdots & 1 & \cdots & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & \cdots & 0 & \cdots & 0 \end{pmatrix},$$

In finite dimension, every linear operator $\hat{A}$ decomposes according to its matrix elements:

$$\hat{A} = \sum_{i,j=1}^n A_{ij} \hat{E}_{ij} = \sum_{i,j=1}^n A_{ij} |e_i\rangle \langle e_j|,$$

where $A_{ij} = \langle e_i | \hat{A} | e_j \rangle$. Caution: in infinite dimension this decomposition is not always possible; and when it does exist, it must be understood in the sense of strong convergence. We will return to this in subsequent lessons.

Resolution of the identity operator. The identity operator in $\mathcal{H}$, denoted $\mathbb{1}$, is defined by $\mathbb{1}|x\rangle = |x\rangle$ for every $|x\rangle \in \mathcal{H}$, trivially. In finite dimension it is, obviously, represented by the identity matrix. More generally, we have the formula

$$\mathbb{1} = \sum_{i \in I} |e_i\rangle \langle e_i| \quad (2.14)$$

also valid in countably infinite dimension, in which case the series converges strongly. It is also called the *resolution of the identity* or the *closure relation*. This formula is extremely useful for all computations in quantum mechanics. Caution: it is false in non-separable spaces².

2.6 The adjoint operator

Let us return to the Riesz isomorphism and consider a continuous linear operator³ $\hat{A}$ from $\mathcal{H}$ to $\mathcal{G}$, where $\mathcal{H}$ and $\mathcal{G}$ are two Hilbert spaces, possibly with $\mathcal{H} = \mathcal{G}$.

When $\hat{A}$ acts on a ket $|u\rangle \in \mathcal{H}$, we obtain a new ket $|v\rangle = |\hat{A}u\rangle = \hat{A}|u\rangle \in \mathcal{G}$. It is natural to ask what the bra associated via the Riesz isomorphism with the vector $|v\rangle$ is, i.e. what the linear form $\langle v| = \langle \hat{A}u|$ on $\mathcal{G}$ is. This will lead us naturally to consider a new operator, called the *adjoint of $\hat{A}$* and likewise denoted $\hat{A}^\dagger$. This operator plays an essential role in quantum mechanics.

²In fact, the decomposition formula $x = \sum_{i \in I} \langle e_i, x \rangle e_i$ remains valid in any Hilbert space, but in the non-separable case the sum runs only over a countable subset of indices $I(x) \subset I$ depending on x . One cannot therefore derive from it a writing of the identity that is independent of the vector on which it is applied, which invalidates formula (2.14) in this setting.

³We restrict here to continuous operators for simplicity. We will return later to the construction of the adjoint in the non-continuous case, cf. the following lesson on the theory of linear operators and the one on unbounded operators.

To determine $\langle v| = \langle \hat{A}u|$, fix any ket $|w\rangle \in \mathcal{G}$ and consider the linear form on $\mathcal{H}$:

$$\begin{aligned} \mathcal{H} &\longrightarrow \mathbb{C} \\ \varphi : |u\rangle &\longmapsto \langle \hat{A}u, w \rangle_{\mathcal{G}} \end{aligned}$$

(Above, the subscript $\mathcal{G}$ indicates in which space the inner product is to be taken.) We will take for granted here that φ is a continuous linear form on $\mathcal{H}$ when $\hat{A}$ is a continuous operator. By the Riesz theorem applied to $\mathcal{H}$, there then exists a unique vector $|z\rangle \in \mathcal{H}$ such that $\varphi = \varphi_z$, that is:

$$\forall |u\rangle \in \mathcal{H}, \quad \varphi(u) = \langle \hat{A}u, w \rangle_{\mathcal{G}} = \varphi_z(u) = \langle u, z \rangle_{\mathcal{H}}.$$

The above construction has thus allowed us to associate with a vector $|w\rangle$ of $\mathcal{G}$ a unique vector $|z\rangle$ of $\mathcal{H}$. This operation can formally be written as the action of an operator, called the adjoint operator and denoted $\hat{A}^\dagger$, in the form $|z\rangle = \hat{A}^\dagger |w\rangle$. Note that the operator $\hat{A}$ acts from $\mathcal{H}$ to $\mathcal{G}$, while its adjoint acts from $\mathcal{G}$ to $\mathcal{H}$.

Completing this construction for every $|w\rangle$, one verifies without difficulty that the operator thus defined is itself a continuous linear operator. We then obtain the following fundamental result:

Definition 2.2 : Adjoint of a continuous operator

The adjoint of a continuous linear operator $\hat{A} : \mathcal{H} \rightarrow \mathcal{G}$ is the unique continuous linear operator $\hat{A}^\dagger : \mathcal{G} \rightarrow \mathcal{H}$ such that:

$$\forall |u\rangle \in \mathcal{H}, \forall |w\rangle \in \mathcal{G}, \quad \langle \hat{A}u | w \rangle_{\mathcal{G}} = \langle u | \hat{A}^\dagger w \rangle_{\mathcal{H}} \quad (2.15)$$

The adjoint has two practical uses in formal quantum calculations.

1. The formula above shows that, so to speak, "with the positions of u and w fixed, the adjoint makes it possible to move the operator from the left-hand side to the right-hand side".
2. Using the Hermitian symmetry of the inner product, $\langle \hat{A}u | w \rangle_{\mathcal{G}} = \langle w | \hat{A}u \rangle_{\mathcal{G}}^*$, we also obtain:

$$\forall |u\rangle \in \mathcal{H}, \forall |w\rangle \in \mathcal{G}, \quad \langle w | \hat{A} | u \rangle_{\mathcal{G}}^* = \langle u | \hat{A}^\dagger | w \rangle_{\mathcal{H}} \quad (2.16)$$

which shows this time that "the adjoint lets one swap bra and ket up to a complex conjugate".

These two identities are equivalent. One must of course pay attention to the source and target spaces $\mathcal{H}$ and $\mathcal{G}$, and to which of the two inner products—that of $\mathcal{H}$ or that of $\mathcal{G}$ —is being used. In practice, however, we will almost always have $\mathcal{H} = \mathcal{G}$, and the notation simplifies accordingly.

Let us now illustrate the above construction in the finite-dimensional case. We have the following main result:

Proposition 2.2 : Conjugate transpose

In finite dimension, the matrix representing the adjoint $\hat{A}^\dagger$ is the conjugate transpose of the matrix representing the operator $\hat{A}$.

$$\boxed{A^\dagger = (A^*)^\top} \quad (2.17)$$

Proof. The matrix elements of $\hat{A}^\dagger$ are by definition:

$$(\hat{A}^\dagger)_{ij} = \langle e_i | \hat{A}^\dagger | e_j \rangle = \langle e_i | \hat{A}^\dagger e_j \rangle.$$

By definition of the adjoint, we have:

$$\langle e_i | \hat{A}^\dagger e_j \rangle = \langle \hat{A} e_i | e_j \rangle$$

and by Hermitian symmetry we have

$$\langle \hat{A} e_i | e_j \rangle = \langle e_j | \hat{A} e_i \rangle^* = \langle e_j | \hat{A} | e_i \rangle^* = A_{ji}^*$$

We have shown:

$$(A^\dagger)_{ij} = A_{ji}^* = (A^{*\top})_{ij}$$

□

We have not yet explicitly answered the question: what is the bra $\langle v | = \langle \hat{A} u |$ canonically associated with the ket $|v\rangle = |\hat{A} u\rangle$? Formula (2.15) answers this via an inner product valid for every vector w , which allows us to write:

$$\forall w \in \mathcal{G}, \quad \langle \hat{A} u | w \rangle_{\mathcal{G}} = \langle u | \hat{A}^\dagger | w \rangle_{\mathcal{H}} \Rightarrow \boxed{\langle \hat{A} u | = \langle u | \hat{A}^\dagger}$$

where the conclusion is obtained by noting that two linear forms that coincide on every vector w are necessarily equal. It is, however, crucial to clearly understand what this object is. Indeed, a common misconception is to infer from this notation that "the adjoint acts on the bras from the left". This is incorrect! The object $\langle u | \hat{A}^\dagger$ is in fact a *composition of operators*, not a left action. Indeed, the source and target spaces are:

$$\begin{aligned} \hat{A}^\dagger &: \mathcal{G} \rightarrow \mathcal{H}, \\ \langle u | &: \mathcal{H} \rightarrow \mathbb{C}, \end{aligned}$$

so that $\langle u | \hat{A}^\dagger$ is the composition $\langle u | \circ \hat{A}^\dagger : \mathcal{G} \rightarrow \mathcal{H} \rightarrow \mathbb{C}$, which is indeed a linear form on $\mathcal{G}$. The $\circ$ symbol is usually omitted, which can be confusing.

Example 2.2 : Illustration in finite dimension

Kets are column vectors, bras are row vectors, and the adjoint is the conjugate transpose. Take, for example:

$$\hat{A} = \begin{pmatrix} 1 & i \\ 0 & 2 \end{pmatrix}, \quad \hat{A}^\dagger = \begin{pmatrix} 1 & 0 \\ -i & 2 \end{pmatrix}, \quad |u\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

We compute:

$$\hat{A}|u\rangle = \begin{pmatrix} 1 & i \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} i \\ 2 \end{pmatrix} \quad \text{from which we deduce} \quad \langle \hat{A}u| = (-i \ 2).$$

Let us verify by direct computation:

$$\langle u|\hat{A}^\dagger = (0 \ 1) \begin{pmatrix} 1 & 0 \\ -i & 2 \end{pmatrix} = (-i \ 2).$$

We indeed recover the same row vector. Here, $\langle u|\hat{A}^\dagger$ is a product of a row vector with a matrix, the result of which is indeed a row vector on $\mathcal{G}$. To convince oneself of the impossibility of a left action, one may try to compute $\hat{A}^\dagger \langle u|$, which matrix-wise reads:

$$\hat{A}^\dagger \langle u| = \begin{pmatrix} 1 & 0 \\ -i & 2 \end{pmatrix} (0 \ 1),$$

which is a product of a 2×2 matrix by a 1×2 row vector: an operation that is meaningless in terms of matrix multiplication.

Finally, from $\langle \hat{A}u| = \langle u|\hat{A}^\dagger$, by taking the dagger we deduce the useful formulas:

$$\left(\hat{A}|u\rangle\right)^\dagger = \langle u|\hat{A}^\dagger, \quad (2.18)$$

$$\left(\langle u|\hat{A}^\dagger\right)^\dagger = \hat{A}|u\rangle. \quad (2.19)$$

Example 2.3 : Computing a formal composition via the adjoint

Let us look at a typical elementary quantum mechanics exam question on the harmonic oscillator. The statement provides a family of kets $|n\rangle$ for positive integers n , and gives the following relations: $\hat{a}|n\rangle = \sqrt{n}|n-1\rangle$ and $\hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$. The question is then asked: what is $\langle n|\hat{a}^\dagger$?

If one believes that $\hat{a}^\dagger$ acts from the left, one is tempted to answer $\langle n|\hat{a}^\dagger = \sqrt{n+1}\langle n+1|$. This answer is clearly wrong, since the equations above instead give, via equation (2.18) with $\hat{A} = \hat{a}$:

$$\langle n|\hat{a}^\dagger = (\hat{a}|n\rangle)^\dagger = (\sqrt{n}|n-1\rangle)^\dagger = \sqrt{n}\langle n-1|$$

The following figure summarizes the relations between bras, kets, the operator $\hat{A}$ and its adjoint seen in this lesson.

We close this section with a few important properties of the adjoint operation:

Proposition 2.3 : Properties of the adjoint operator

For all continuous linear operators $\hat{A}, \hat{B}$ and every scalar $\lambda \in \mathbb{C}$:

$$(\hat{A} + \hat{B})^\dagger = \hat{A}^\dagger + \hat{B}^\dagger, \quad (2.20)$$

$$(\lambda \hat{A})^\dagger = \lambda^* \hat{A}^\dagger, \quad (\text{antilinearity}) \quad (2.21)$$

$$(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger \hat{A}^\dagger, \quad (\text{watch out for the order}) \quad (2.22)$$

$$(\hat{A}^\dagger)^\dagger = \hat{A}, \quad (\text{involution}). \quad (2.23)$$

Proof. (composition case): for all $|u\rangle \in \mathcal{H}$ and $|w\rangle \in \mathcal{G}$, we apply the fundamental identity Eq. (2.15) twice:

$$\langle (\hat{A}\hat{B})u | w \rangle = \langle \hat{A}(\hat{B}u) | w \rangle = \langle \hat{B}u | \hat{A}^\dagger w \rangle = \langle u | \hat{B}^\dagger (\hat{A}^\dagger w) \rangle = \langle u | (\hat{B}^\dagger \hat{A}^\dagger) w \rangle.$$

We conclude by uniqueness of the adjoint. □

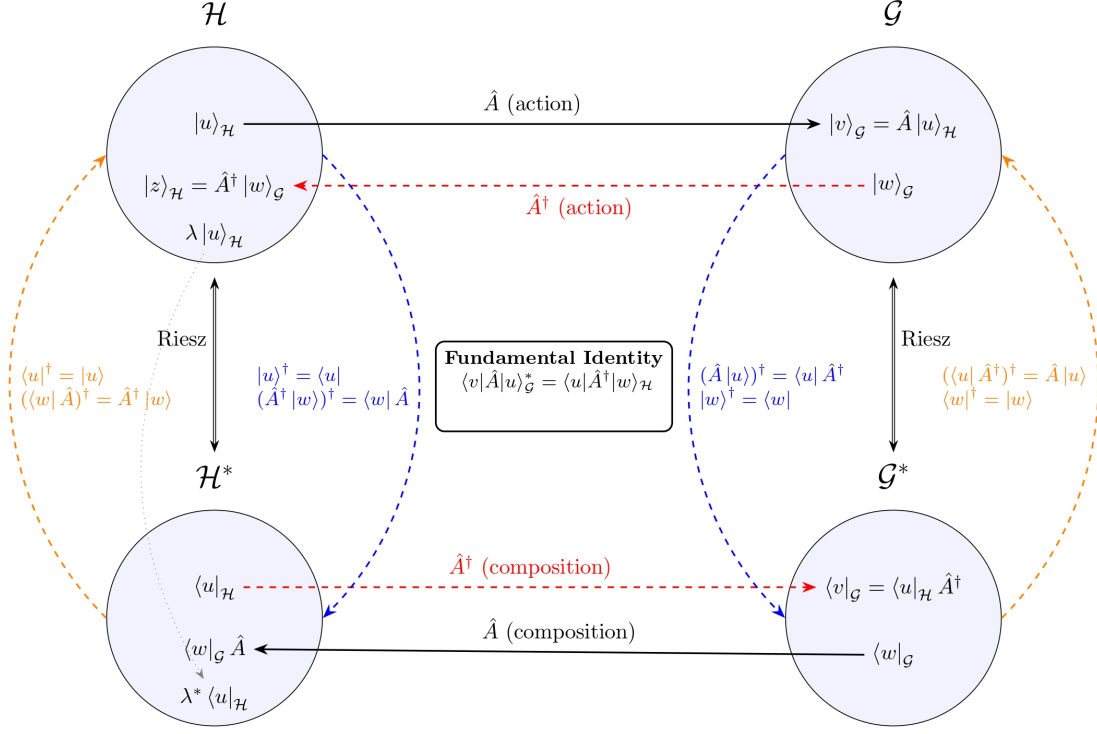


Figure 2.1: Representation of the source and target Hilbert spaces $\mathcal{H}$ and $\mathcal{G}$, together with their topological duals $\mathcal{H}^*$ and $\mathcal{G}^*$. The operator $\hat{A}$ has a natural right action: it transforms a ket in the source space $|u\rangle_{\mathcal{H}}$ into a ket in the target space $\hat{A}|u\rangle_{\mathcal{H}} \in \mathcal{G}$. The Riesz isomorphism enables one to construct its adjoint $\hat{A}^\dagger$, which acts naturally on kets from $\mathcal{G}$ to $\mathcal{H}$ in such a way as to satisfy *the fundamental identity*. Caution: the diagram should not suggest that this is the inverse map! In general, $\hat{A}^\dagger \neq \hat{A}^{-1}$. The upper horizontal arrows represent operator actions, while the lower horizontal arrows represent compositions as discussed in the text. The blue and orange arrows indicate the dagger operation (Riesz isomorphism Φ or its inverse Φ^{-1}). The text specifies the essential formulas for these operations (where we have omitted the subscripts $\mathcal{H}$ and $\mathcal{G}$ to keep things readable). Since this operation is involutive, the blue and orange transformations are inverses of each other. The dashed gray line recalls the antilinearity of this isomorphism, which is the source of the complex conjugate in the fundamental identity. The subscripts $\mathcal{H}$ and $\mathcal{G}$ in this identity specify in which space each inner product is computed.